

Arsen Khanguieldyan

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AI engineer with 7 years shipping production AI/ML systems across autonomous robotics, IoT, and enterprise data. Hands-on with Agentic RAG, LLM tool-use, computer vision, and the full deploy spine (FastAPI, Docker, GitHub Actions, Terraform, AWS, Azure). Recent focus: building and operationalizing LLM agents on AWS Bedrock-class stacks.

SELECTED PROJECTS

Site Copilot — Agentic RFI & Daily Report Assistant

2026

suffolk-construction.up.railway.app · github.com/smartpneucontact-sketch/Sufflk

- Designed and shipped two production-shaped LLM agents on Claude Sonnet 4.6 with Agentic RAG over a synthetic construction project corpus; tool-use loop portable to AWS Bedrock Converse with no code change.
- Hybrid retrieval (BM25 + dense vectors, fused via Reciprocal Rank Fusion) behind an OpenSearch-shaped VectorStore Protocol; FastAPI + Pydantic service, Docker image, GitHub Actions CI gating on a YAML eval suite.
- LLM Ops spine: JSONL tracing (run_id, latency, tokens, cost) shaped for Databricks bronze, rolling drift detector, Terraform sketch (ECS Fargate + OpenSearch + Bedrock IAM).
- End-to-end RFI triage runs at ~\$0.05 per call against a 27-chunk corpus, producing cited responses with urgency, multi-scenario cost/schedule impact, and EOR-escalation flag.

PROFESSIONAL EXPERIENCE

Hyperion

Nov 2022 – Jun 2025

Head of Engineering · Yerevan, Armenia

- Shipped a fully autonomous AI defense drone — acquired by strategic buyer; owned engineering across AI software, computer vision, firmware, hardware, and composites.
- Built edge-deployed computer vision for GPS-denied navigation at 96% accuracy and target engagement at 90% reliability under active jamming and hacking; validated across 3,000 flight-test hours.
- Led a multidisciplinary team (AI, CV, firmware, electronics, mechanical); trained, validated, and deployed CV models to on-device inference.
- Designed custom PCB in Altium for flight control and ESC; ran control-loop tuning in Simulink with hardware-in-the-loop validation.
- Developed a proprietary composite for structural components, reducing airframe weight while maintaining structural performance.

Deloitte

Apr 2021 – Jun 2022

Data Analyst · Luxembourg

- Migrated multi-source enterprise data to a unified repository on Azure with static and dynamic metadata; exposed it via REST API to a single-page web app.
- Evaluated nine AutoML platforms (IaaS / PaaS / SaaS) against predefined test protocols; produced selection rationale for executive review.

Forschungsgesellschaft Umformtechnik mbH

May 2019 – Mar 2021

Data Engineer · Stuttgart, Germany

- Shipped a tool-wear classifier (CNN, 82% accuracy) integrated into TRUMPF Group's next-generation punching machines; co-authored "Data-Driven Tool Wear Classification with a Convolutional Neural Network in Punching Machines" (Feb 2020).
- Built a sensor-fusion IoT pipeline for stamping presses (force, distance, sound, lubrication); developed an ML failure-prediction algorithm on Audi AG data.

Audi AG

Apr 2018 – Dec 2018

Software Developer · Neckarsulm, Germany

- Built and deployed an Oracle APEX web application digitalizing die-cast tooling improvements; rolled out internationally across Audi sites.

TECHNICAL SKILLS

AI & ML

LLMs (Claude, GPT), Agentic RAG, tool-use, prompt engineering, PyTorch, HuggingFace, YOLO, computer vision, CUDA, LLMOps

Backend & Cloud

Python (expert), FastAPI, REST, C/C++, SQL · AWS (Bedrock-class), Azure (AZ-400, AZ-900, AZ-204), Docker, Kubernetes, Terraform, GitHub Actions, Databricks-shaped pipelines, ETL

Data & Retrieval

Vector stores (Chroma), hybrid search (BM25 + dense, RRF), OpenSearch-shaped interfaces, Azure Data Factory

Electronics & Hardware

Altium / PCB design, Arduino, sensor integration, SolidWorks, Simulink

CERTIFICATIONS

AWS Cloud Practitioner · TensorFlow Developer · Deep Learning in Computer Vision · Stochastic Processes

EDUCATION

Ecole Centrale d'Electronique — Paris, France

M.S. Computer Science & Engineering Jun 2019

B.S. Mathematics & Electronics Jun 2017

MIT Professional Education

Dec 2025 – Sep 2026

Digital Transformation in the AI Age (in progress)

LANGUAGES

English (proficient) · French (fluent) · German (fluent) · Armenian (native)